
STATISTICAL MACHINE LEARNING FOR DATA SCIENCE

ASSIGNMENT 2

Due Date: Sunday, December 15, 2024 - By 11:59pm

Exercise 1: Practical SML on DNA Microarrays (60 points)

Consider the prostate cancer dataset containing the DNA MicroArray Gene Expression of both cancer and non-cancer subjects.

```
library(class)
library(rpart)
library(rpart.plot)
library(ROCR)
```

```
prostate <- read.csv('prostate-cancer-1.csv') # DNA MicroArray Gene Expression
```

You are supposed to provide a thorough comparison of six learning machines on this dataset, namely 1NN, 7NN, 9NN, `Trees (cp=0)`, `Trees (cp=0.05)`, and `Trees (cp=0.1)`, and your comparison will be solely based on the test error.

```
prostate$Y <- as.factor(prostate$Y)
tree.xy <- rpart(Y~., data=prostate, control = rpart.control(cp = 0.01))
```

1. Comment on the shape of this dataset in terms of the sample size and the dimensionality of the input space
2. Comment succinctly from the statistical perspective on the type of data in the input space
3. Plot the distribution of the response for this dataset and comment.
4. Identify the 9 individually most powerful predictor variables with respect to the response according the Kruskal-Wallis test statistic
5. Generate a `type='h'` plot with the Kruskal-Wallis test statistic as the y-axis and the variable name as the x-axis
6. Generate the comparative boxplots of the 9 most powerful variable with respect to the response and comment on what you observe.
7. Build the classification tree with `cp=0.01`
 - Plot the tree you just built

- Determine the number of terminal nodes
 - Write down in mathematical form region 2 and Region 4.
 - Comment on the variable at the root of the tree in light of the Kruskal-Wallis statistic
8. Generate the comparative boxplots of the 9 weakest variable with respect to the response and comment on what you observe.
 9. Generate the correlation plot of the predictor variables and comment extensively on what they reveal, if anything.
 10. Compute the eigendecomposition of the correlation matrix and comment on the ratio $\lambda_{max}/\lambda_{min}$.
 11. Using the whole data for training and the whole data for test, build the above six learning machines, then plot the comparative ROC curves on the same grid
 12. Plot all the three classification tree grown, using the `prp` function for the package `rpart.plot`
 13. Comment succinctly on what the ROC curves reveal for this data and argue in light of theory whether or not that was to be expected.
 14. Using `set.seed(19671210)` along with a 7/10 training 3/10 test basic stochastic holdout split of the data, compute $S = 100$ replicated random splits of the test error for all the above learning machines.
 - Plot the comparative boxplots (be sure to properly label the plots)
 - Comment on the distribution of the test error in light of (implicit) model complexity.
 - Perform a basic analysis of variance (ANOVA) on those test errors and comment!
 15. Comment extensively on the most general observation and lesson you gleaned from this exploration.

Exercise 2: Nearest Neighbors Method for Digit Recognition (30 points)

This exercise features the analysis of the USPS digit recognition dataset using kNN with various neighborhood sizes.

```
library(dslabs)          # Package by Yann LeCun to provide the MNIST data
mnist <- read_mnist()    # Read in the MNIST data
xtrain <- mnist$train$images
ytrain <- mnist$train$labels
ytrain <- as.factor(ytrain)
ntr   <- nrow(xtrain)
p     <- ncol(xtrain)
xtest <- mnist$test$images
ytest <- mnist$test$labels
ytest <- as.factor(ytest)
```

Part 1: Multi-class classification on MNIST

Throughout this part of the exercise, you will perform multiclass classification in the MNIST data using the learning machines 1NN, 5NN, 7NN, 9NN, and 13NN.

1. Write down in mathematical form the expression of $\hat{f}_{\text{kNN}}(\mathbf{x})$, the prediction function of the kNN learning machine.
2. Choose n a training set size and m a test set size, and write a piece of code for sampling a fragment from the large dataset. Explain why you choose the numbers you chose.
3. Let $S = 50$ be the number of random splits of the data into 70% training and 30% Test.
 1. Build over all the 5 models and compute the test errors for each split, storing the results into a matrix of test errors
 2. Identify the machine with the smallest median test error and generate the test confusion matrix from the last split
 3. Comment on the digits for which there is a lot more confusion. Does that agree with your own prior intuition about digits?
4. Perform an ANOVA of the test errors and comment on the patterns that emerge.

Part 2: Binary classification on MNIST

Consider classifying digit '1' against digit '7', with '1' representing positive and '7' representing negative. You will be using just 1NN, 5NN, 7NN, 9NN, and 13NN.

1. Store in memory your training set and your test set. Of course you must show the command that extracts only '1' and '7' from both the training and the test sets.
2. Display both your training confusion matrix and your test confusion matrix
3. Display the comparative ROC curves of the five learning machines
4. Identify two false positives and two false negatives at the test phase, and in each case, plot the true image against its falsely predicted counterpart.
5. Comment on any pattern that might have emerged.

Exercise 3: Video component (10 points)

Every student must complete a 5 minutes (Max 5 minutes) video exactly as was for the first homework assignment! You are still strongly encouraged to make as much use of Artificial Intelligence as possible, but mindless copies of LLM outputs will result in loss of points or even a zero for the corresponding question.